

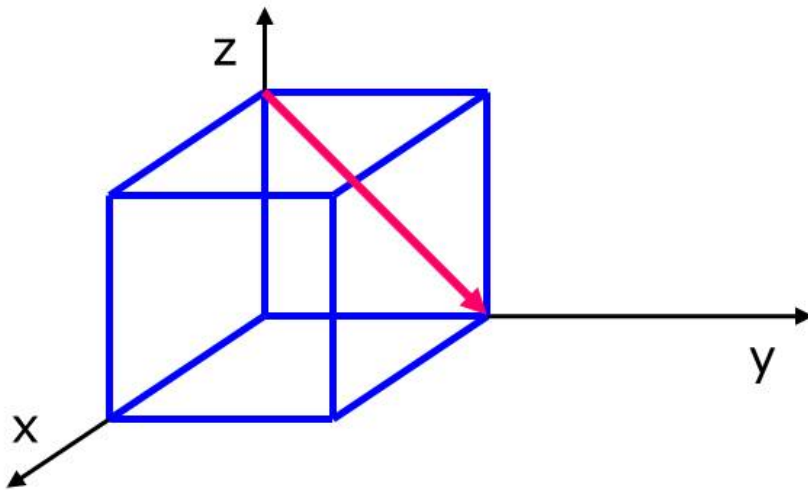
CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Crystal structures and bonds

Question 1

What is the correct crystallographic notation for the crystallographic direction indicated in the unit cell?

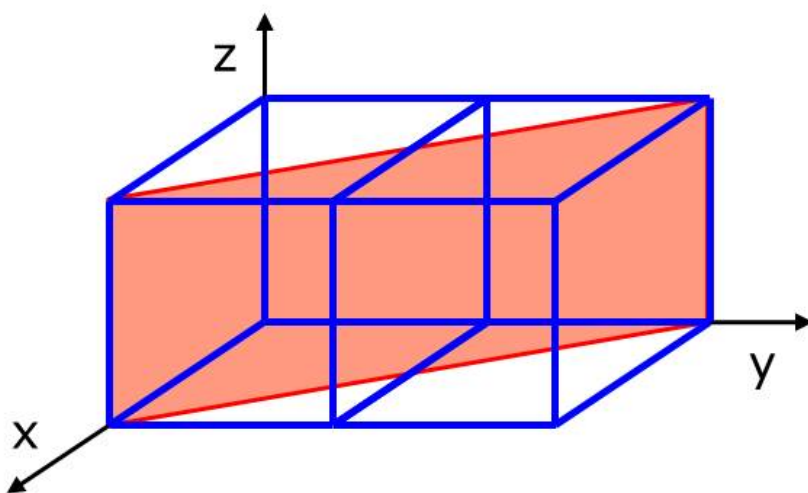


- ☐ $\langle 0\bar{1}1 \rangle$
- ☐ $[0\bar{1}1]$
- ☐ $\langle 01\bar{1} \rangle$
- ☒ $[01\bar{1}]$



Question 2

What is the correct crystallographic notation for the crystallographic plane indicated in the unit cell(s)?



- ☒ (210)
- ☐ $(2\bar{1}0)$
- ☒ (120)
- ☐ $(\bar{1}20)$



Question 3

Which is the closest packed direction in a body-centered cubic (bcc) lattice?

- ☐ $\langle 100 \rangle$
- ☐ $\langle 110 \rangle$
- ☒ $\langle 111 \rangle$ ✓
- ☐ None of the above

Question 4

Which types of interatomic bonding occur in ceramics?

- ☐ ionic bonds only
- ☐ covalent bonds only
- ☒ ionic as well as covalent bonds ✓
- ☐ ionic, covalent as well as metallic bonds

Question 5

What are the crystal structures of the two salts, potassium iodide (KI) and caesium iodide (CsI)?

element	Cs	K	I
valence	+1	+1	-1
electronegativity	0.9	0.9	2.2
atomic radius / pm	260	220	140
ionic radius / pm	170	138	220

- ☐ both salts have ZnS structure
- ☐ both salts have NaCl structure
- ☐ both salts have CsCl structure
- ☒ both salts have different crystal structures ✓

[Go to front-page](#)

[Save and go to next](#)

[Jump to](#)



CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Lattice defects

Question 6

Which lattice defects are present in thermal equilibrium?

- ☒ vacancies ✓
- ☐ dislocations
- ☐ grain boundaries
- ☐ voids

Question 7

Which of the following statements concerning dislocations is wrong?

- ☐ Dislocations are the origin of elastic stresses.
- ☐ The motion of dislocations causes plastic deformation.
- ☐ The dislocation density increases during plastic deformation.
- ☒ Dislocations are planar defects. ✓

Question 8

Where are foreign atoms not expected, when a pure metal is alloyed with a second element of slightly larger atomic size?

- ☒ on regular lattice sites ✗  (✓)
- ☐ on interstitial sites ✓
- ☐ at dislocations
- ☐ at grain boundaries

Question 9

Compare the face-centered cubic (fcc) lattice with the body-centered cubic (bcc) lattice.

- ☐ bcc is denser packed and contains only smaller interstitial sites than fcc
- ☒ fcc is denser packed and contains only smaller interstitial sites than bcc ✗ 
- ☐ bcc is denser packed and contains larger interstitial sites than fcc
- ☐ fcc is denser packed and contains larger interstitial sites than bcc ✓

Previous Save and go to next Jump to 

CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Miscibility / Immiscibility

Silver (Ag) shall be alloyed with either copper (Cu) or gold (Au). Ag and Au are completely miscible in the solid state and form a solid solution, whereas Ag and Cu are completely immiscible in the solid state.

Element	Cu	Ag	Au
Electronegativity	1.8	1.4	1.4
Atomic radius / pm	128	145	144
Density / (g/cm ³)	8.9	10.5	19.3
Young's modulus / GPa	130	83	78
Yield stress / MPa	25	45	25

Question 10

What is the Young's modulus for a binary AgCu alloy with 20 vol.% Cu and 80 vol.% Ag?

- ☐ below 83 GPa
- ☐ between 83 GPa and 89 GPa
- ☒ between 89 GPa and 93 GPa ✓
- ☐ between 93 GPa and 130 GPa

Question 11

Can the yield strength of silver be increased by alloying with either 0.5 wt.% Cu or 0.5 wt.% Au?

- ☐ Yes, but only by alloying with Cu.
- ☒ Yes, but only by alloying with Au. ✓
- ☐ Yes, by alloying with either Cu or Au.
- ☐ No, the yield strength cannot be increased, neither by alloying with Cu, nor with Au.

Question 12

For complete miscibility in the solid state, two elements must, in general, have ...

- ☐ ... similar electrical conductivity
- ☐ ... similar melting temperature
- ☐ ... similar mass density
- ☒ ... similar lattice constant ✓

Previous Save and go to next Jump to ↓

CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Polymers

Question 13

Polymers have a rather small mass density due to a combination of reasons. Which of the following properties is not relevant for the low mass density?

- ☐ Polymers consist mainly of elements with low atomic number.
- ☐ Atoms in a polymer are bond by short covalent bonds. ✓ 
- ☐ Atoms in a polymer have only a few neighboring atoms next to them.
- ☒ Between individual polymer chains only weak van der Waals forces occur. ✗

Question 14

What is the origin of the extremely low Young's modulus of elastomers (in particular before vulcanisation)?

- ☒ covalent bonds within the polymer chains ✗
- ☐ van der Waals force between the polymer chains
- ☐ hydrogen bonds between specific side groups
- ☐ none of the above ✓ 

Question 15

Which characteristic type of bonds distinguishes elastomers and thermosets from thermoplastics?

- ☐ van der Waals forces
- ☐ hydrogen bonds
- ☐ ionic bonds
- ☒ covalent crosslinks ✓

Question 16

Which statement is true for thermoplastics, but wrong for elastomers and thermosets?

- ☐ They can melt. ✓ 
- ☒ They can be a glass. ✗
- ☐ They can be semi-crystalline.
- ☐ They can be crystalline.

Previous

Save and go to next

Jump to



CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Mechanical properties

Question 17

Which mechanical property is not affected by dislocations?

- ☒ Young's modulus ✓
- ☐ Yield strength
- ☐ Tensile strength
- ☐ Fracture toughness

Question 18

How can metals be strengthened?

- ☐ by increasing the vacancy concentration
- ☐ by increasing the impurity concentration ✓ 
- ☒ by increasing the grain size ✗
- ☐ by increasing the density of voids

Question 19

Mechanical properties, as stiffness or ductility, depend on temperature. When lowering the temperature of body-centered cubic metals below room temperature ...

- ☐ both, their stiffness and ductility, increase
- ☒ their stiffness increases, whereas the ductility decreases ✓
- ☐ their stiffness decreases, whereas their ductility increases
- ☐ both, their stiffness and ductility, decrease

Question 20

Caesium chloride (CsCl) and β' -brass (a CuZn alloy with 50 at.% Cu and 50 at.% Zn) have the same crystal structure at room temperature, but their mechanical behavior is quite different: CsCl is brittle, whereas CuZn behaves ductile. What is the main reason for this behavior?

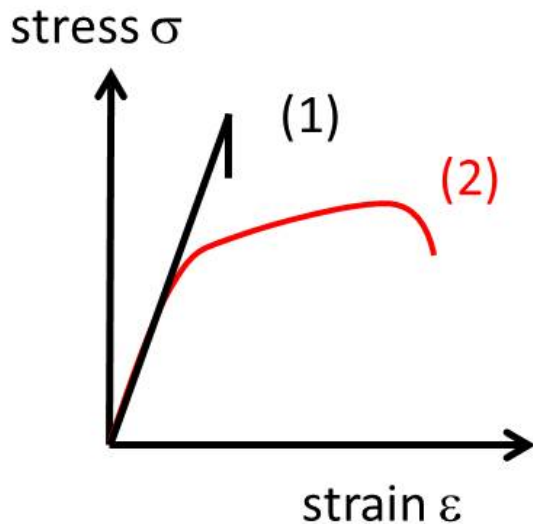
- ☐ the different melting temperature
- ☐ the difference in the atomic size between the constituting elements
- ☒ the difference in the type of bonding between the constituting elements ✓
- ☐ the difference in the molar mass of the constituting elements

CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Stress strain curves

The following stress strain curves (1) and (2) shall be discussed.



Question 21

Comparing the two different stress strain curves, ...

- ☐ (1) has lower tensile strength and higher ductility than (2)
- ☐ (1) has lower tensile strength and lower ductility than (2)
- ☐ (1) has higher tensile strength and higher ductility than (2)
- ☒ (1) has higher tensile strength and lower ductility than (2)



Question 22

Is it possible that both flow curves are measured for specimens with exactly the same chemical composition?

- ☐ Yes, because they have exactly the same Young's modulus ✓
- ☐ Yes, if the flow curves are measured at different temperatures
- ☒ No, because one specimen shows brittle, the other ductile behavior ✗
- ☐ No, because one specimen shows work-hardening, whereas the other does not deform plastically

Previous Save and go to next Jump to ▾

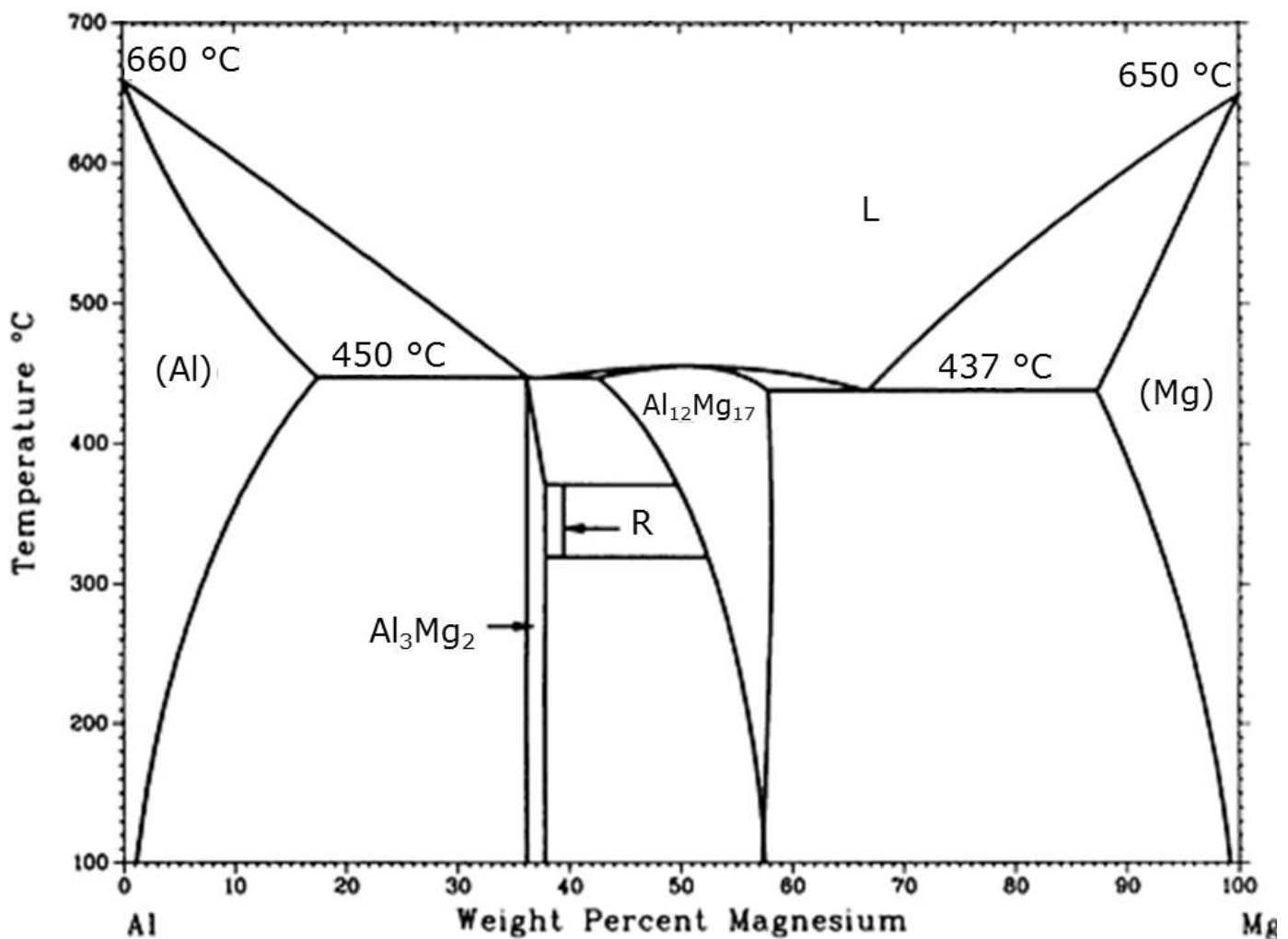
CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Phase diagram (I)

A molten aluminium magnesium alloy containing 10 wt.% Mg is slowly cooled from 700 °C under conditions of thermal equilibrium.

The figures shows the phase diagram of the binary system aluminium-magnesium (Al-Mg).



Question 23

At which temperature starts solidification of an AlMg alloy with 10 wt.% Mg?

- ☐ at 450 °C
- ☐ at 520 °C
- ☒ at 600 °C
- ☐ at 660 °C



Question 24

What is first solid phase to be formed during solidification of an AlMg alloy with 10 wt.% Mg?

- ☒ (Al) with 4 wt.% Mg
- ☐ (Al) + L with 4 wt.% Mg
- ☐ (Al) with 10 wt.% Mg
- ☐ (Al) + L with 10 wt.% Mg



Question 25

What is the maximum magnesium content in the melt during solidification of an AlMg alloy with 10 wt.% Mg?

- ☐ exactly 10 wt.%
- ☐ about 25 wt.%
- ☒ about 36 wt.%
- ☐ above 36 wt.%



Question 26

Two phases will be present after equilibrium cooling of an AlMg alloy to 100 °C. One is Al₃Mg₂ with 36 wt.% Mg, what is the other phase?

- ☒ (Al) with 1 wt.% Mg
- ☐ (Al) with 4 wt.% Mg
- ☐ (Al) with 10 wt.% Mg
- ☐ (Al) with 17 wt.% Mg



Question 27

What is the maximum solubility of aluminium in solid magnesium?

- ☒ about 12 wt.%
- ☐ about 99 wt.%
- ☐ about 88 wt.%
- ☐ about 1 wt.%



Previous

Save and go to next

Jump to

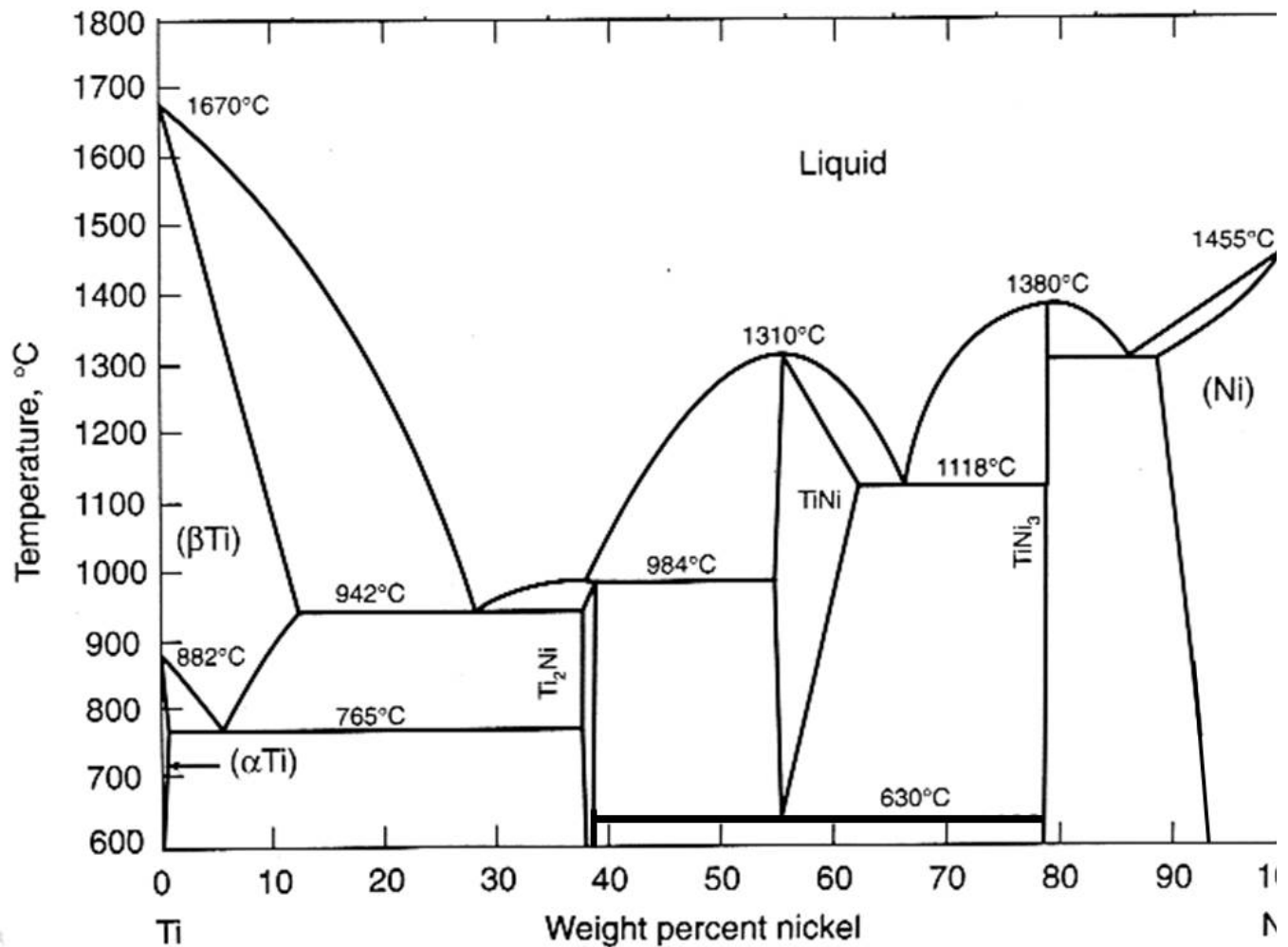


CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Phase diagram (II)

The figure shows the phase diagram of the binary system titanium-nickel (Ti-Ni).

**Question 28**

What is the maximum solubility of nickel in solid titanium?

- ☐ less than 1 wt. %
- ☐ about 5 wt. %
- ☒ about 12 wt. %
- ☐ about 88 wt. %

**Question 29**

Is there an eutectic point present in the phase diagram?

- ☐ Yes, at 765 °C for 5 wt. % Ni
- ☒ Yes, at 1118 °C for 66 wt. % Ni
- ☐ Yes, at 1380 °C for 78 wt. % Ni
- ☐ No, there is no eutectic point

**Question 30**

What is the lowest temperature, for which a melt of titanium and nickel can exist in thermal equilibrium?

- ☐ 765 °C
- ☒ 942 °C
- ☐ 1118 °C
- ☐ 1310 °C



Question 31

Which phases exist at 800 °C for an titanium nickel alloy with 20 wt.% Ni in thermal equilibrium?

- ☐ One phase, (β -Ti) + Ti₂Ni
- ☐ Two phases, (α -Ti) and Ti₂Ni
- ☒ Two phases, (β -Ti) and Ti₂Ni
- ☐ Three phases, (α -Ti), (β -Ti) and Ti₂Ni



Question 32

How much nickel solid solution (Ni) does a specimen with mass 5 kg of a titanium nickel alloy with 82 wt.% Ni contain at 600 °C (in thermal equilibrium)?

- ☐ 5 kg
- ☐ More than 2.5 kg
- ☒ Some, but less than 2.5 kg
- ☐ None



Previous

Save and go to next

Jump to



Exam from F17**Image analysis****Question 33**

What properties does Gaussian filtering have?

- ☐ It will preserve edges, i.e. the abrupt changes in intensity in the image.
- ☐ It makes the image sharper, but amplifies noise.
- ☒ It is good for removing noise, but suppresses edges. ✓
- ☐ It will make the intensity histogram of the image follow a normal distribution.

Question 34

An image contains the following pixel values:

63	81	141	195	195	177
74	97	155	183	189	177
82	111	160	185	178	167
79	101	149	175	180	171
75	92	147	190	190	171
66	100	135	186	191	191

The image should be filtered using a filter with the values:

1/16	2/16	1/16
2/16	4/16	2/16
1/16	2/16	1/16

What is the pixel value of the filtered image in coordinate (3,4), if the output is represented in double precision?
Note, that the input image has a pixel value of 185 in coordinate (3,4).

- ☐ 146.25
- ☐ 175
- ☒ 175.3125 ✓
- ☐ 185

Question 35

Fibers are identified in an image (as in the exercise) by first Gaussian filtering the image and then finding the location of local maxima. If one views the image as a 2D continuous signal, one would find the location of local maxima where the derivative is zero, which in 2D means the location where the partial derivatives in the x- and y- directions are zero. How does one find the location of maxima in a discrete image?

- ☒ The location of maxima can be found in a manner similar to a continuous 2D signal, but for discrete data one must find maxima where the partial derivatives change sign. In practice, the easiest is to compare the pixel value to its neighbors and find the location where the pixel value is larger than its neighbors. ✓
- ☐ In practice, it is difficult to find the location of local maxima in an image, because the image is discretized in pixels.
- ☐ The location of local maxima can only be found by approximating the discrete 2D image using a local approximation of a 2D signal and determining the location where the partial derivatives in the x- and y- directions are zero.
- ☐ The pixel in a local maximum can have a value that is lower than one of the neighbors, that the pixel is compared to for finding the maximum.

Question 36

A scanner gives an image with minimum pixel value 3218 and maximum pixel value 4612. Transform the data to 8 bit unsigned integer utilizing the full dynamic range. What will the pixel value 3416 be transformed to?

☒ 36 ✓

☐ 198

☐ 255

☐ 3514

Previous

Save and go to next

Jump to ▾

CampusNet / 41680 Introduction to cyber materials F18

Exam from F17

Transport phenomena

Question 37

Which of the following modifications has the least effect on the electrical resistivity of pure, polycrystalline copper?

- ☐ alloying with a second element
- ☒ deforming solely elastically
- ☐ deforming plastically
- ☐ raising the temperature



Question 38

At room temperature, gold (Au) has a face-centered cubic (fcc) crystal structure with a lattice constant of 406 pm. Assume an electrical conductivity of $4.3 \cdot 10^7 \text{ (W m)}^{-1}$ and six free electrons per unit cell. What will be their drift velocity for an electrical field of 12 V/m?

- ☐ 12 m/s
- ☐ 5.7 m/s
- ☐ 0.18 m/s
- ☒ 0.036 m/s



Question 39

What is characteristic for the electron energy band scheme of arsenic-doped silicon (As-doped Si)?

- ☐ The induced holes cause an energy band without energy gap.
- ☒ There is an energy gap, with acceptor states close to the conduction band.
- ☐ There is an energy gap, with donor states close to the conduction band.
- ☐ The donor electrons cause an energy band without energy gap.



Question 40

The hole and electron mobilities of InP are $0.015 \text{ m}^2/(\text{Vs})$ and $0.46 \text{ m}^2/(\text{Vs})$, respectively. The electrical conductivity is $2.5 \cdot 10^{-6} \text{ (W m)}^{-1}$. What is the carrier concentration for InP at room temperature?

- ☒ $3.28 \cdot 10^{13} \text{ m}^{-3}$
- ☐ $3.39 \cdot 10^{13} \text{ m}^{-3}$
- ☐ $3.50 \cdot 10^{13} \text{ m}^{-3}$
- ☐ $1.04 \cdot 10^{15} \text{ m}^{-3}$

